

Extraction, Transformation and Load

Knowledge Objectives

1. Enumerate six reasons to have an ETL process
2. Explain some legal and ethical concerns
3. Define ETL
4. Compare ETL, ELT and ETQ
5. Enumerate three levels of data profiling
6. Enumerate six mechanisms to extract data
7. Enumerate five kinds of transformation tasks
8. Enumerate three criteria to select the sources
9. Explain the two integration problems
10. Explain the three kinds of data cleaning activities
11. Justify the reduction of the data size
12. Explain two possibilities to reduce the size of data
13. Enumerate six kinds of preparation activities
14. Discuss the three abstraction levels of ETL process design
15. Discuss the implementation alternatives of ETL flows
16. Enumerate four criteria of ETL process quality
17. Discuss the differences between Data and Control flow in ETL
18. Justify the existence of a staging area
19. Discuss five kinds of ETL metadata
20. Enumerate the different ETL operation
21. Distinguish between blocking and non-blocking operations

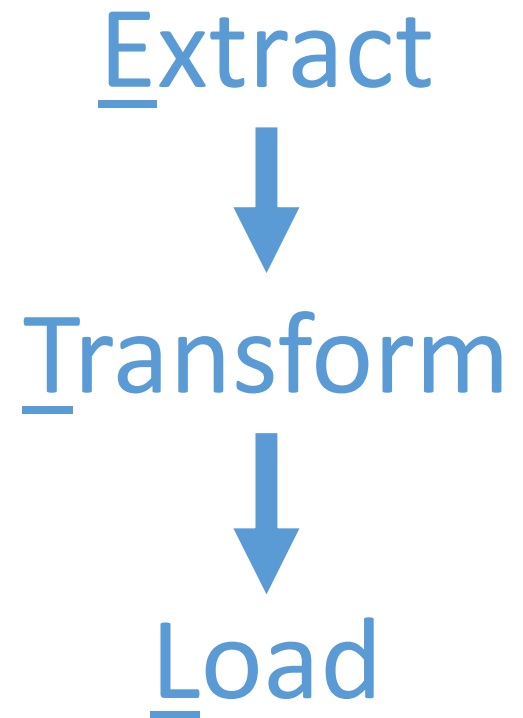
Application Objectives

1. Given 2-3 sources, a target and some requirements create the corresponding ETL flow

Definition and Motivation

Motivation

- Transactional data safekeeping
- Restructure data to be used by other tools
- Use multiple sources together
- Provide measures of confidence in data
- Remove mistakes
- Correct and impute missing data



Legal and ethical aspects

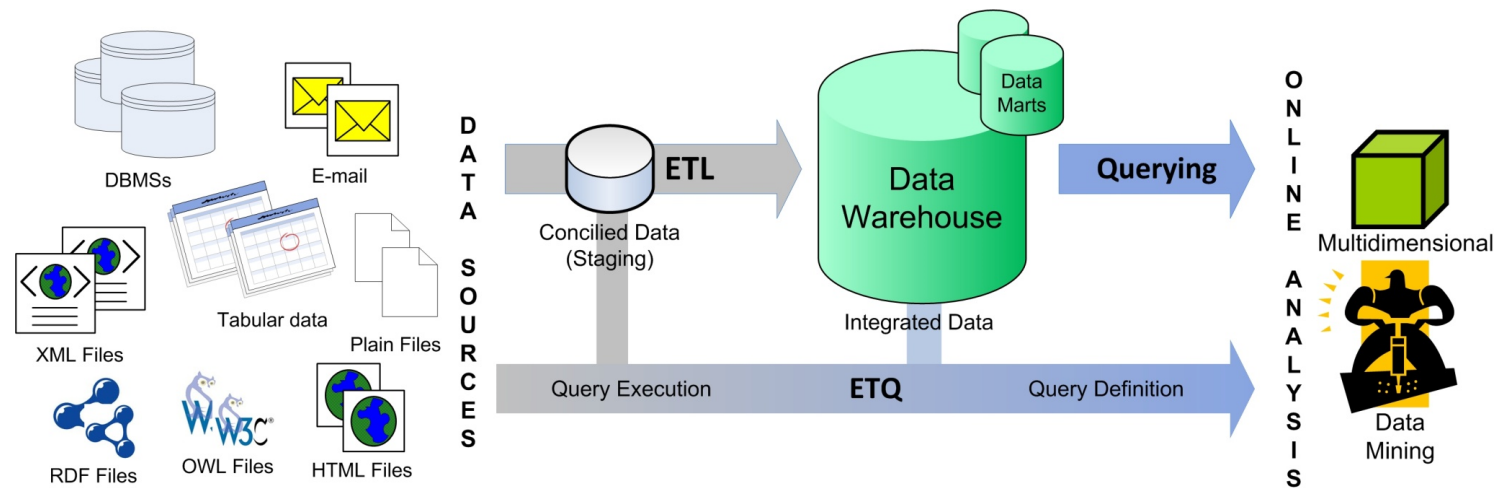
- Who is the owner?
- Do we have permission ...
 - ... To use them with our aim?
- Will we keep the confidentiality?
- How will the results be used?



Definition and characteristics

- Extract
 - Multiple and heterogeneous sources
 - Different temporal characteristics of sources:
 - Transient
 - Semi-periodic
 - Temporal
- Transform
 - Change schema
 - Convert characters set
 - Cleaning
- Load
 - On-line/Off-line (i.e., update window)
 - Update/Rebuild indexes

ETL flows

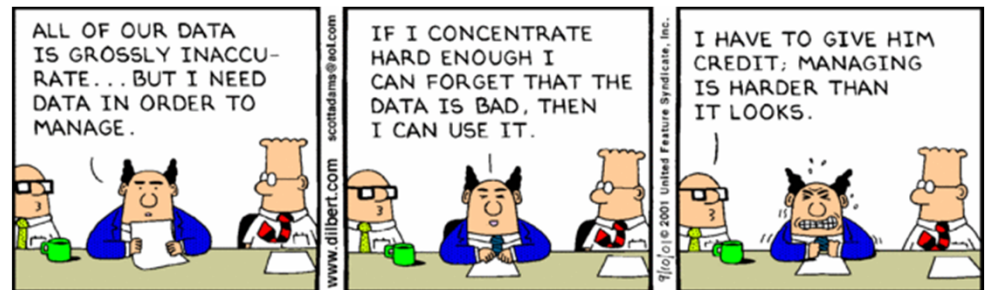


ETL Steps

Source selection

- Starting from the high-level business objectives and target MD schema
 - Relevant
 - Non-redundant
 - High quality
 - Complete
 - Accurate
 - Consistent
 - Timely
- Data profiling

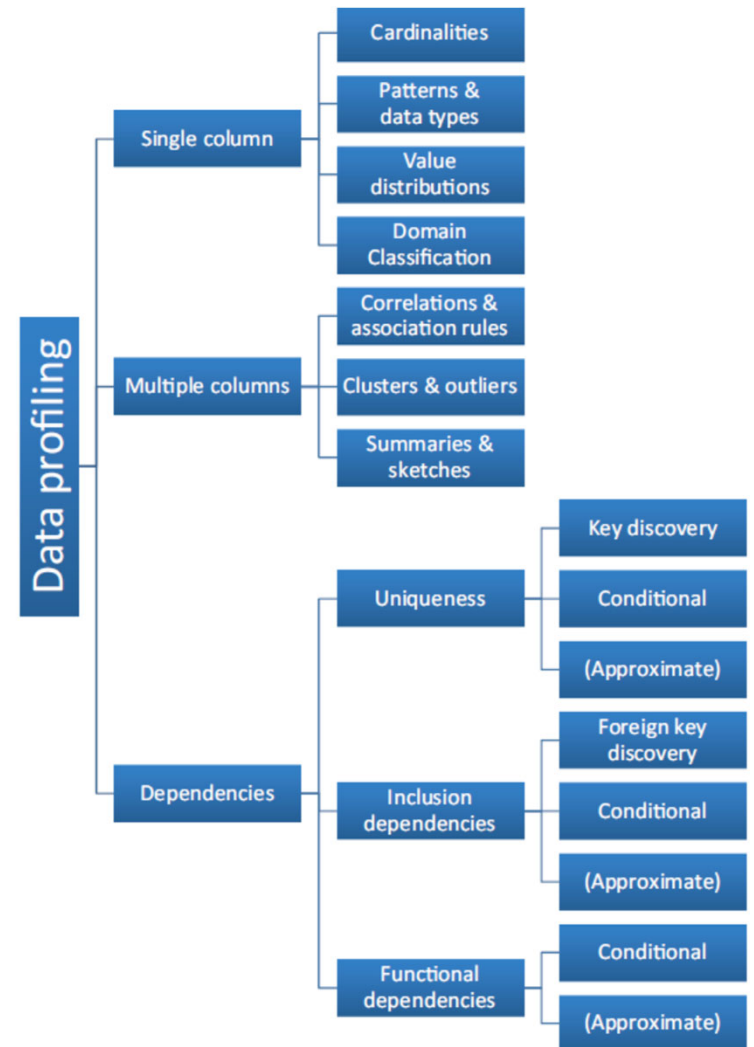
Garbage in-Garbage out



<https://www.wearetheliving.com/myth-garbage-garbage>

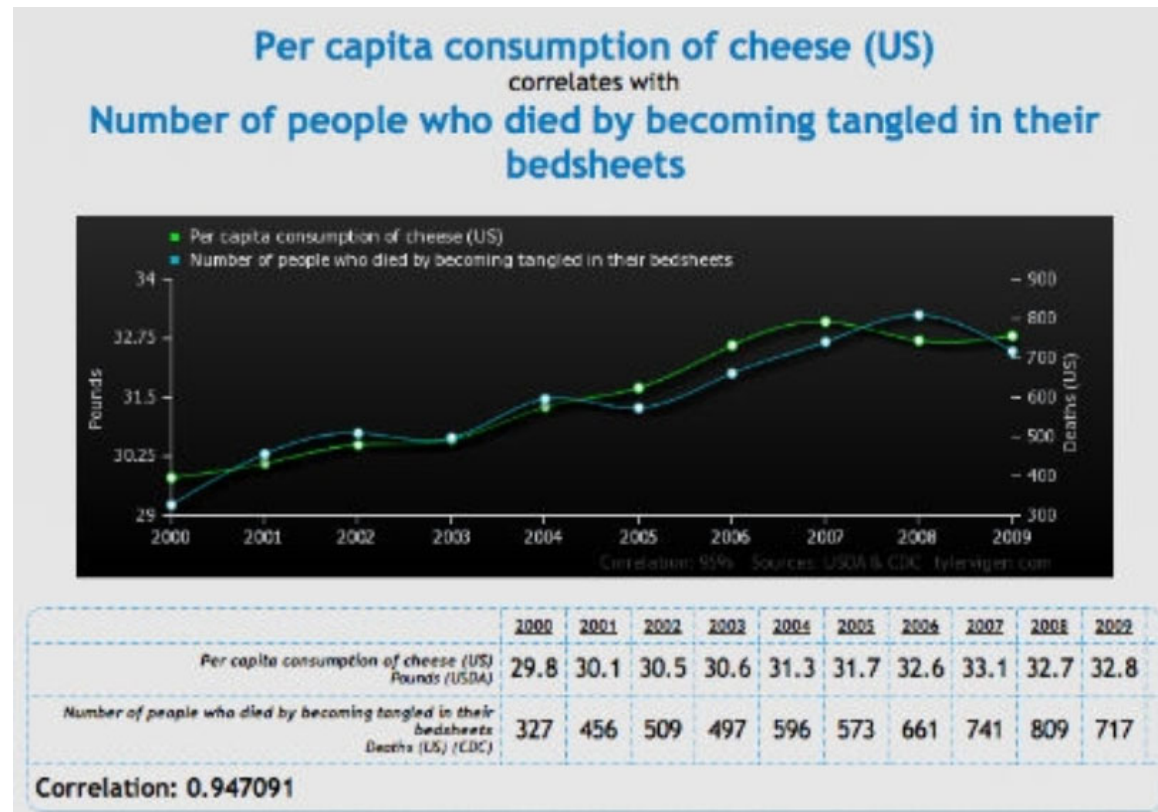
Source selection: data profiling

- Selected source data profiled for
 - Data quality
 - Data characteristics
 - Fitness for the purpose



Z. Abedjan et al. VLDB 2015

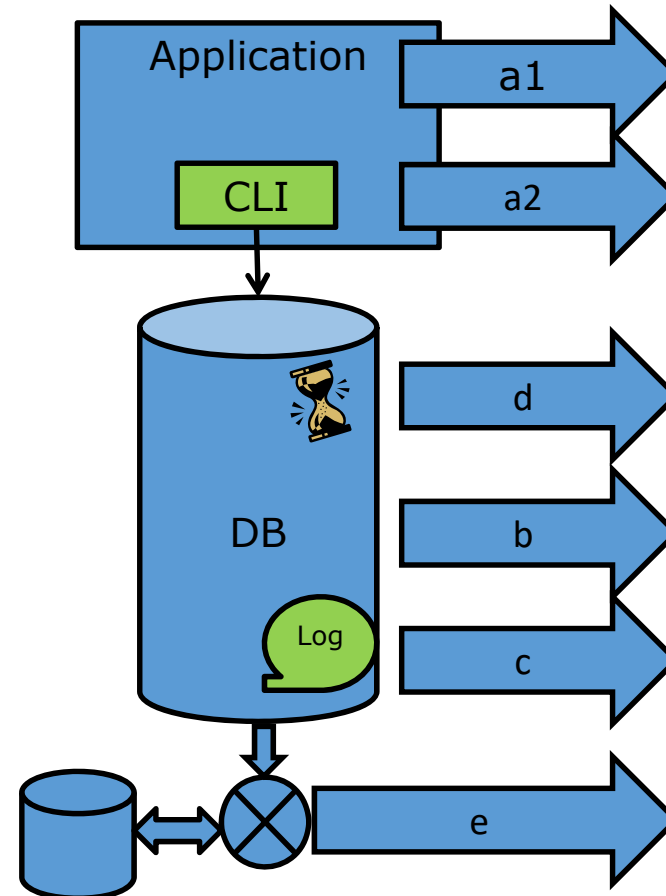
Source selection: data relevance example



https://dangerousminds.net/comments/spurious_correlations_between_nicolas_cage_movies_and_swimming_pool

Data extraction mechanisms

- a) Application-assisted
- b) Trigger-based
- c) Log-based
- d) Timestamp-based
- e) File comparison



Kinds of transforming tasks

“The process of standardizing data representation and eliminating errors in data.”

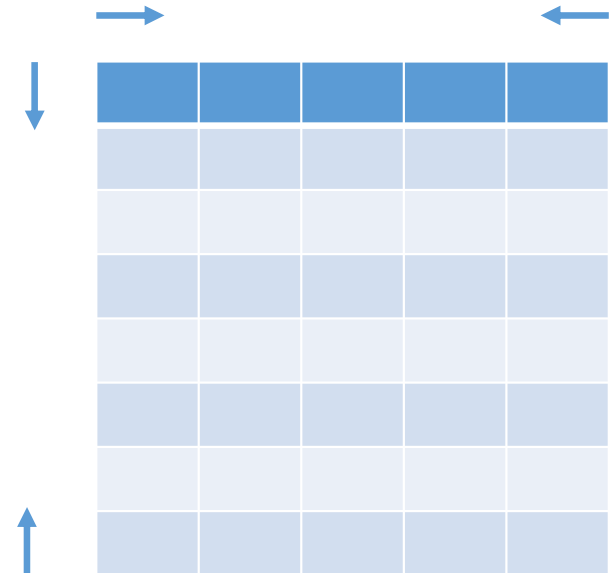
- Selection
- Cleaning
- Reduction
- Preparation
- Integration

Data cleaning

- Generate data profiles
- Segment (split) columns
- Standardize
- Improve quality
 - Complete
 - Introduce values manually
 - Introduce a default value
 - Provide the average/median/mode
 - Provide the average/median/mode of the class
 - Provide the value contributing with more information
 - Cross multiple sources (use a lookup table)
 - Correct
 - Match dictionary (or lookup table)
 - Detect outliers
 - Variance analysis
 - High distance to the regression function
 - High distance to any cluster
 - Check constraints and business rules

Size reduction

- Length (i.e., remove tuples)
 - Aggregate
 - Find a representative (i.e., clustering)
 - Sampling (keeping outliers)
- Width (i.e., remove attributes)
 - Correlation
 - Analysis of significance
 - Information gain (w.r.t. a classification)



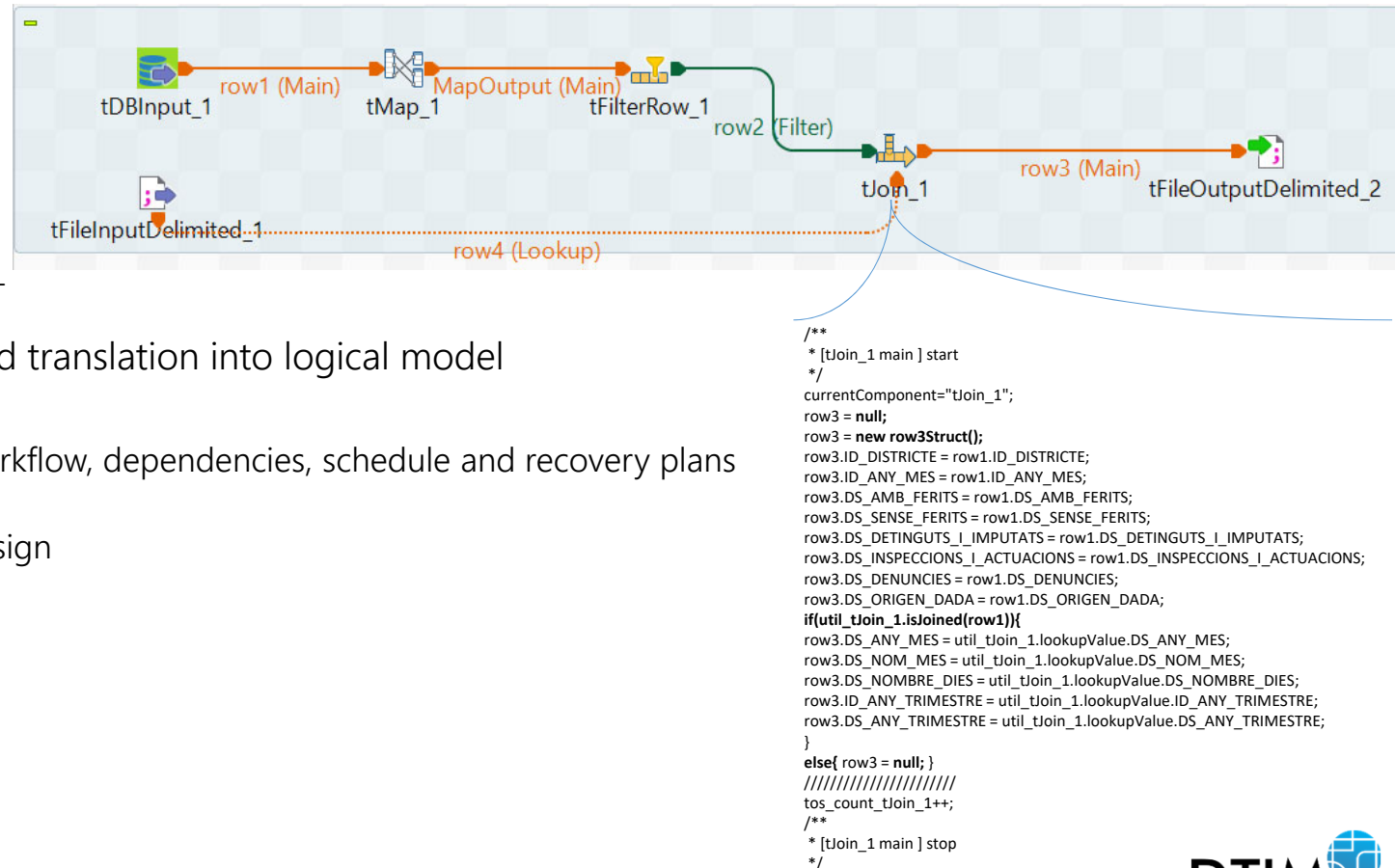
Preparation

- Categorical to numerical
- Numerical to categorical (i.e., discretization)
 - Intervals of the same size
 - Intervals of the same provability
 - Clustering
 - Based on entropy
- Normalization
 - By the maximum ($x/\max$)
 - By the interval size ($|x-\min|/|\max-\min|$)
 - By the standard deviation ($(x-\mu)/\sigma$)
 - Scaling ($x/10^j$)
- Conversion of data into metadata (e.g., one-hot encoding)
- Derivation
- Enrichment (i.e., joins)

ETL Process Design

Design abstraction levels

- Conceptual design
 - Purpose:
 - Understanding
 - Sharing
 - Documenting
 - Existing formalizations
 - Standards like BPMN or UML
 - Domain-specific languages
 - Possible (partial) automated translation into logical model
- Logical design
 - Detailed description of the workflow, dependencies, schedule and recovery plans
 - Directed Acyclic Graph (DAG)
 - Drag&drop manual logical design
 - ETL tools (GUI)
- Physical design
 - Automatic code generation
 - Template-based
 - Hand-coded ETL



ETL tools (GUI)

“The goal of a valuable tool is not to make trivial problems mundane, but to make impossible problems possible.”

- Recommended for
 - Large projects
 - Sophisticated processing
 - Limited programming skills
- Extra features:
 - Integrated metadata repository
 - Handle complex data type conversions
 - Handle complex dependencies and error handling
 - Automatic data lineage and dependency analysis
- Examples
 - Kettle-Pentaho Data Integration
 - cloverETL
 - JasperETL
 - Talend
 - Informatica

Hand-coded ETL

- Recommended when:
 - Know-how already available
 - Reuse legacy routines
 - Not limited to vendor's abilities
- MapReduce-like coding is specially appropriate
 - Schema-less data
 - "Read once" data sets
 - "Cooking" raw data ...
 - ... and loading them in a DBMS
 - Complex data flow

ETL process quality

- Performance
 - Time efficiency
 - Scalability
 - Resource utilization
- Reliability
 - Robustness (deals with errors)
 - Fault tolerance (continue in the presence of errors)
 - Recoverability (recover from failures)
- Auditability
 - Testability
 - Traceability
 - Example: “% of KPIs that can be followed, discovered or ascertained by end users”
- Maintainability

Maintainability

- Overlooking it can cause problems and higher costs
- Mitigating actions
 - Understandable and documented design
 - Conceptual and logical ETL design
- Not easily quantifiable
 - Potential metrics
 - Size (e.g., #operations, #sources)
 - Modularity/Encapsulation

Architectural Setting

Data flow vs. control flow

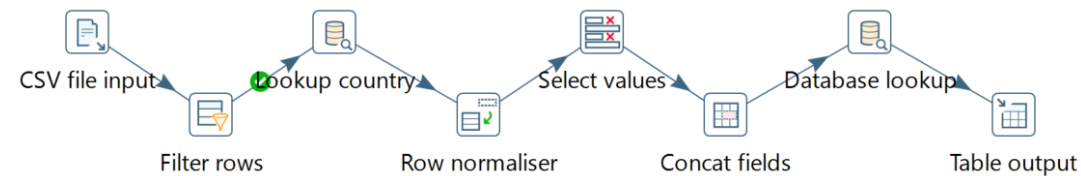
Staging area

Metadata management

Data flow vs. control flow

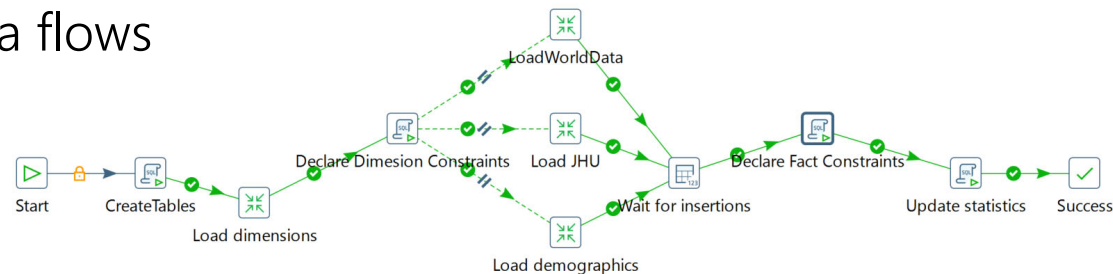
- Data Flows (a.k.a. Transformations)

- Transformation over data
- Execution order
 - Pipelined data
 - No strict order of execution of alternative paths



- Control Flows (a.k.a. Jobs)

- Orchestrating the execution of data flows
- Do not directly work with data
- Execution in strictly defined order
 - Can be sequential or parallel
- Important for dependent data processing
 - Example: dimension tables -> fact tables



Staging area

- Storage only for ETL purposes
- Structures
 - Plain files
 - XML
 - Relational tables
- Provides
 - Recoverability
 - Backup
 - Auditing

Metadata management

- Source system metadata
- Data-Staging metadata
- Target system (e.g., DW) metadata
- Process execution metadata
- Business rules
 - Business metadata
 - Technical metadata

ETL Operations

ETL operations (I)

Operation Level	Operation Type	Pentaho PDI	Talend Data Integration	SSIS	Oracle Warehouse Builder
Field	Field Value Alteration	Add constant Formula Number ranges Add sequence Calculator Add a checksum	tMap tConvertType tReplaceList	Character Map Derived Column Copy Column Data Conversion	Constant Operator Expression Operator Data Generator Transformation Mapping Sequence
Dataset	Duplicate Removal	Unique Rows Unique Rows (HashSet)	tUniqRow	Fuzzy Grouping	Deduplicator
	Sort	Sort Rows	tSortRow	Sort	Sorter
	Sampling	Reservoir Sampling Sample Rows	tSampleRow	Percentage Sampling Row Sampling	
	Aggregation	Group by Memory Group by	tAggregateRow tAggregateSortedRow	Aggregate	Aggregator
	Dataset Copy		tReplicate	Multicast	
Row	Duplicate Row	Clone Row	tRowGenerator		
	Filter	Filter Rows Data Validator	tFilterRow tMap tSchemaComplianceCheck	Conditional Split	Filter
	Join	Merge Join Stream Lookup Database lookup Merge Rows Multiway Merge Join Fuzzy Match	tJoin tFuzzyMatch	Merge Join Fuzzy Lookup	Joiner Key Lookup Operator
	Router	Switch/Case	tMap	Conditional Split	Splitter
	Set Operation - Intersect	Merge Rows (diff)	tMap	Merge Join	Set Operation
	Set Operation - Difference	Merge Rows (diff)	tMap		Set Operation
	Set Operation - Union	Sorted MergeAppend streams	tUnite	Merge Union All	Set Operation

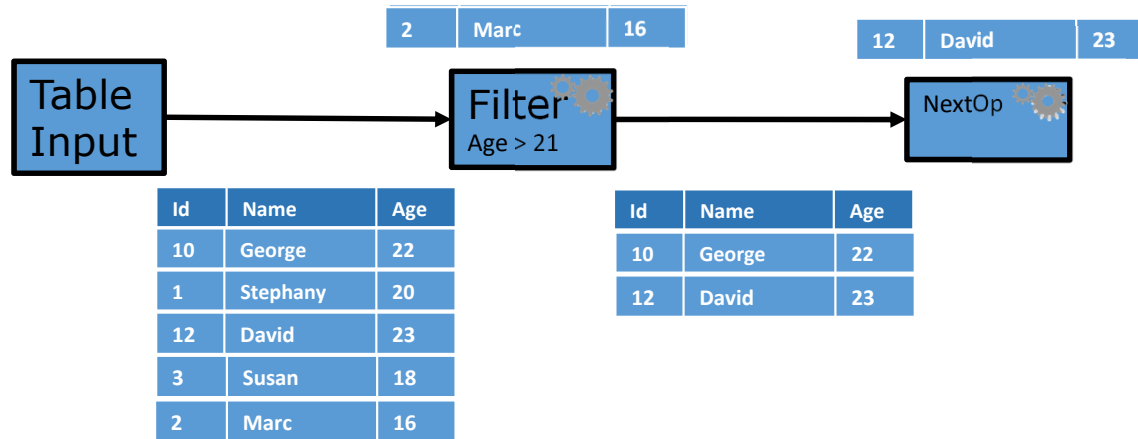
Vasileios Theodorou

ETL operations (II)

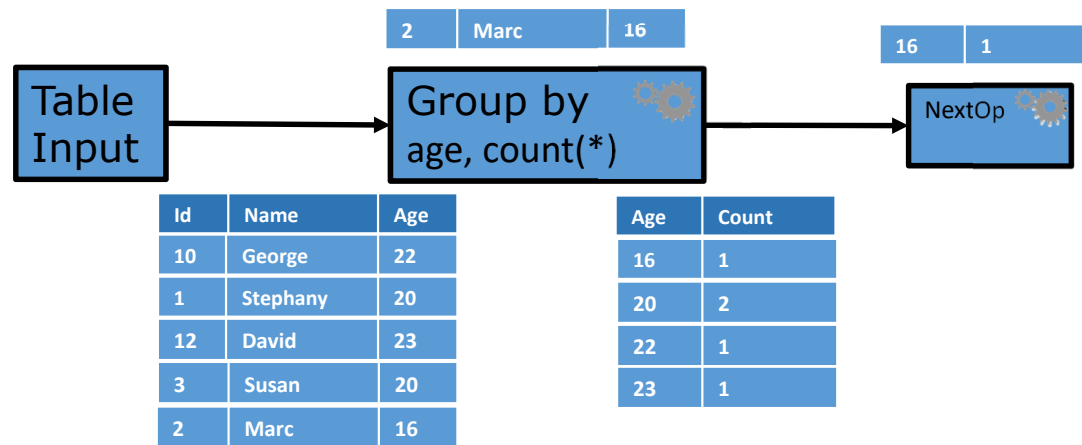
Operation Level	Operation Type	Pentaho PDI	Talend Data Integration	SSIS	Oracle Warehouse Builder
Schema	Field Addition	Set field value Set field value to a constant String operations Strings cut Replace in string Formula Split Fields Concat Fields Add value fields changing sequence Sample rows	tMap tExtractRegexFields tAddCRCRow	Derived Column Character Map Row Count Audit Transformation	Constant Operator Expression Operator Data Generator Mapping Input/Output parameter
	Datatype Conversion	Select Values	tConvertType	Data Conversion	Anydata Cast Operator
	Field Renaming	Select Values	tMap	Derived Column	
	Projection	Select Values	tFilterColumns		
Table	Pivoting	Row Denormalizer	tDenormalize tDenormalizeSortedRow	Pivot	Unpivot
	Unpivoting	Row Normalizer Split field to rows	tNormalize tSplitRow	Unpivot	Pivot
Value	Single Value Alteration	If field value is null Null if Modified Java Script Value SQL Execute	tMap tReplace	Derived Column	Constant Operator Expression Operator Match-Merge Operator Mapping Input/Output parameter
Source Operation	Extraction	CSV file input Microsoft Excel Input Table input Text file input XML Input	tFileInputDelimited tDBInput tFileInputExcel	ADO .NET / DataReader Source Excel Source Flat File Source OLE DB Source XML Source	Table Operator Flat File Operator Dimension Operator Cube Operator
Target Operation	Loading	Text file output Microsoft Excel Output Table output Text file output XML Output	tFileOutput tDelimited tDBOutput tFileOutputExcel	Dimension Processing Excel Destination Flat File Destination OLE DB Destination SQL Server Destination	Table Operator Flat File Operator Dimension Operator Cube Operator

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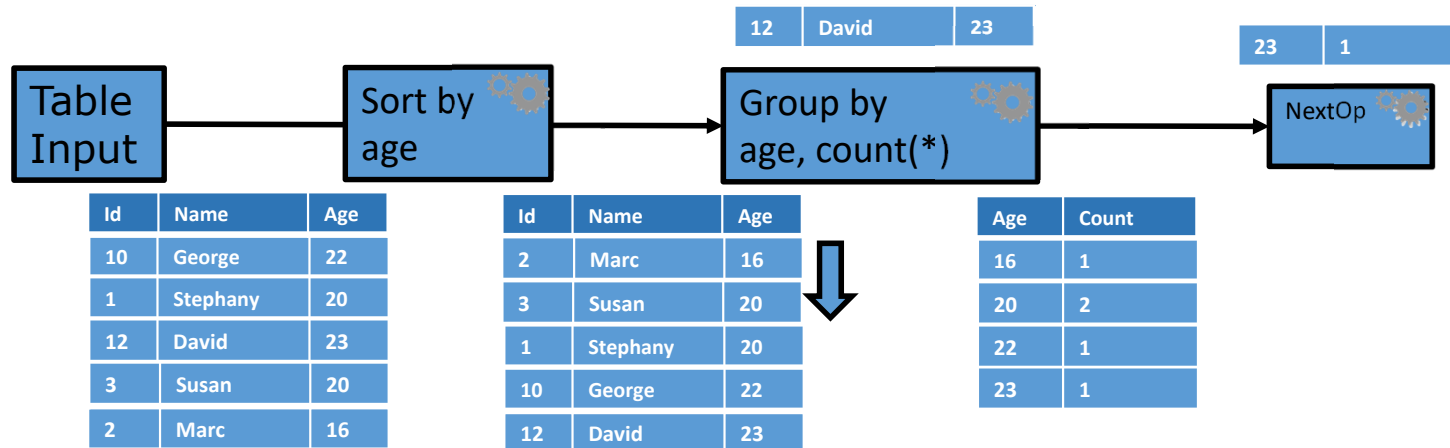
Non-blocking operation (example)



Blocking operation (example)



Blocking operation - optimized (example)



Kinds of operations

- Blocking
 - Duplicate Removal
 - Sort
 - Aggregation
 - Join
 - Intersect
 - Difference
 - Unpivot (transform rows into columns)
- Non-blocking
 - Field Value Alteration
 - Single Value Alteration
 - Sampling
 - Dataset Copy
 - Duplicate Row
 - Router
 - Union
 - Field Addition
 - Datatype Conversion
 - Field Renaming
 - Projection
 - Pivot (transform columns into rows)
 - Extraction
 - Loading

Closing

Summary

- Extraction, Transformation and Load
- Kinds of transformation tasks
 - Selection
 - Integration
 - Cleaning
 - Reduction
 - Preparation
- ETL process design
- ETL process quality
- Architectural setting
 - Data flow vs. Control flow
 - Staging area
 - Metadata management
- Operations
 - Blocking vs. non-blocking

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